

RTD-Lite: Scalable Topological Analysis for Comparing Weighted Graphs in Learning Tasks

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Introduction

Topological methods for comparing weighted graphs are valuable in various learning tasks but often suffer from computational inefficiency on large datasets. We introduce RTD-Lite, a scalable algorithm that efficiently compares topological features, specifically connectivity or cluster structures at arbitrary scales, of two weighted graphs with one-to-one correspondence between vertices. Using minimal spanning trees in auxiliary graphs, RTD-Lite captures topological discrepancies with $O(n^2)$ time and memory complexity.

Experiments demonstrate that RTD-Lite

- effectively identifies topological differences between graphs;
- has significantly reduced computation time compared to competitive methods;
- enhances the preservation of topological structures in learned representations

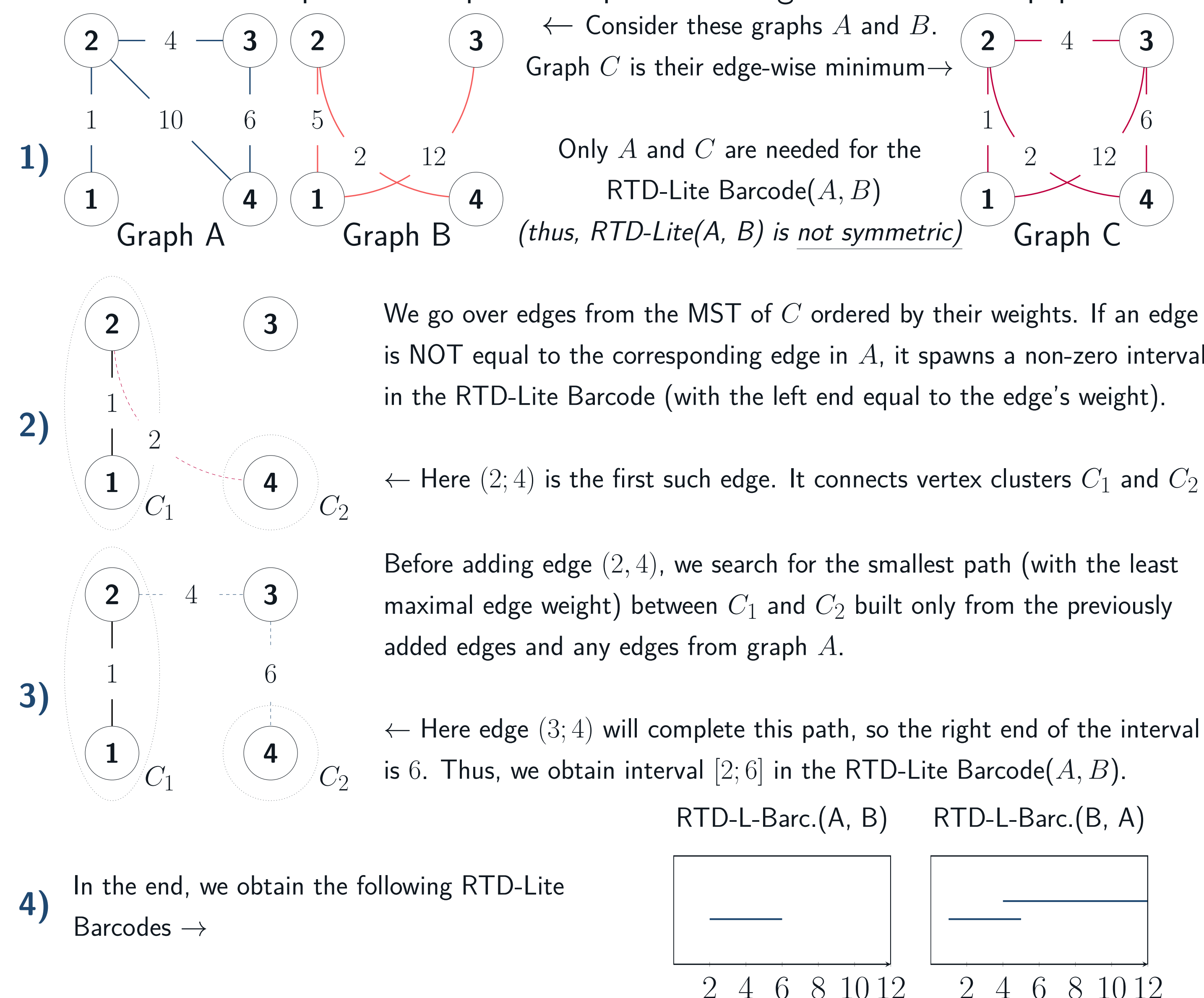
Our code is publicly available at github.com/ArGintum/RTD-Lite

RTD-Lite

Let A and B be two connected, undirected, weighted graphs on n vertices, with a bijection between their vertex sets (so vertex i in A corresponds to vertex i in B).

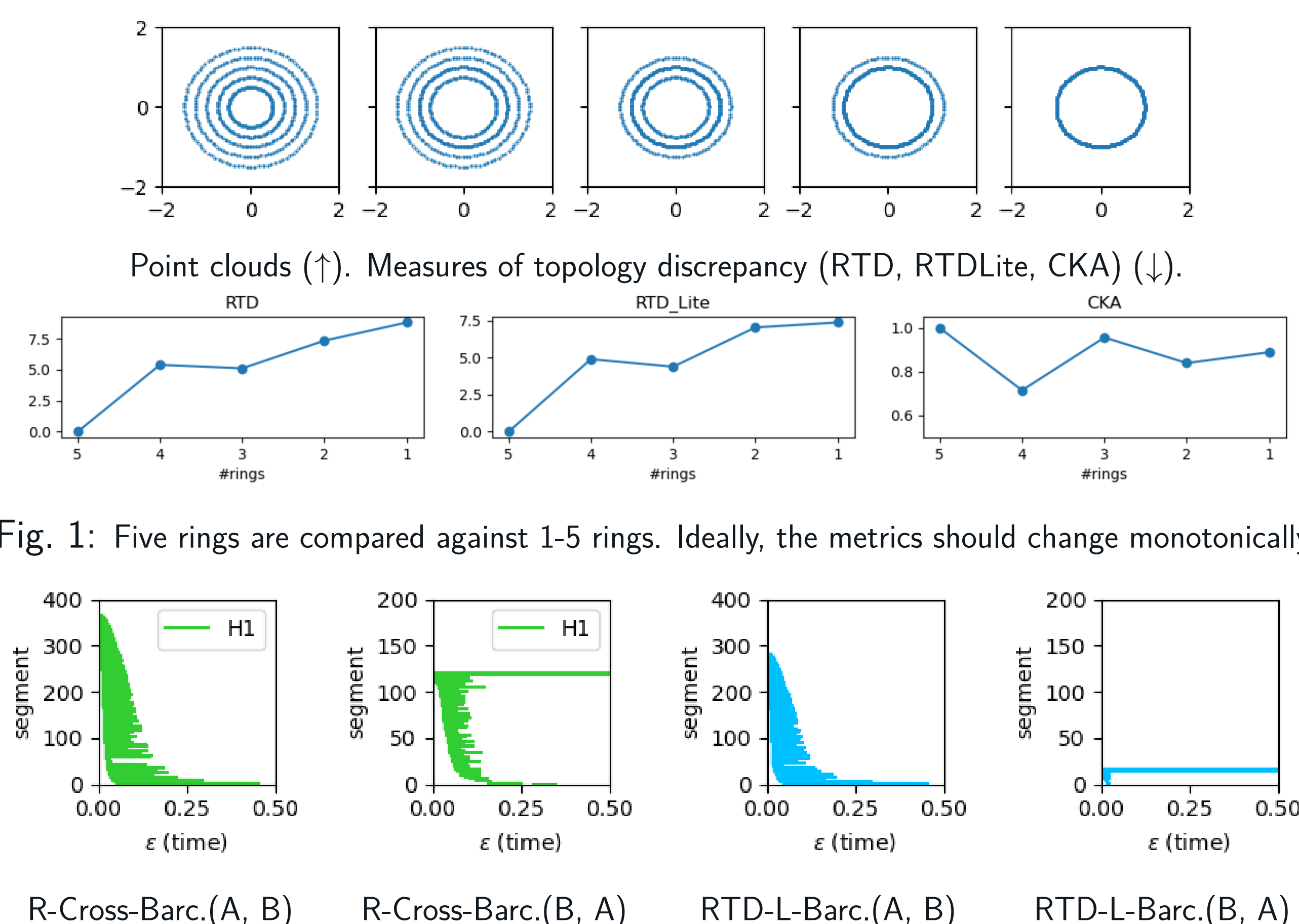
► **RTD-Lite Barcode** is a collection of intervals, representing misalignments in multi-scale cluster structure of the two weighted graphs.

Here, we give a simple example on how intervals from RTD-Lite Barcode are obtained. For the detailed description of the procedure please see Algorithm 1 in the paper.



► **RTD_Lite (RTDL(A, B))** is the sum of length of intervals from the RTD-Lite Barcode(A, B). It can be computed from MSTs of A and C via simplified scheme (Algorithm 2) with $O(n^2)$ time complexity, although other statistics of RTD-Lite Barcode will require direct computations (Algorithm 1) with $O(n^2\alpha(n))$ time complexity.

More examples



Comparing DNN Internal Representations

Convolutional Neural Networks (CNN)

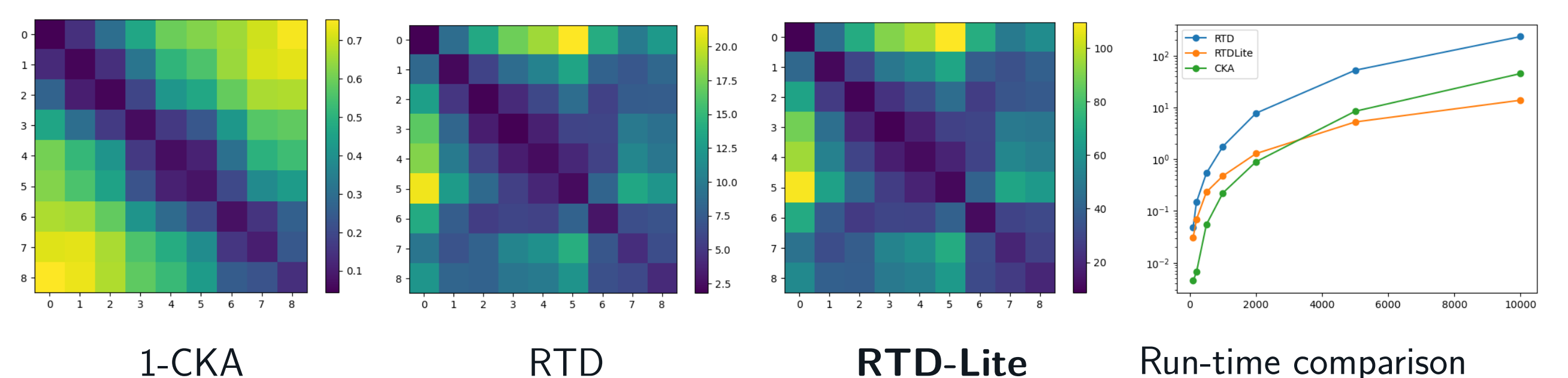


Fig. 3: Comparing layers from All-CNN trained on CIFAR10 with different seeds.

Graph Neural Networks (GNN)

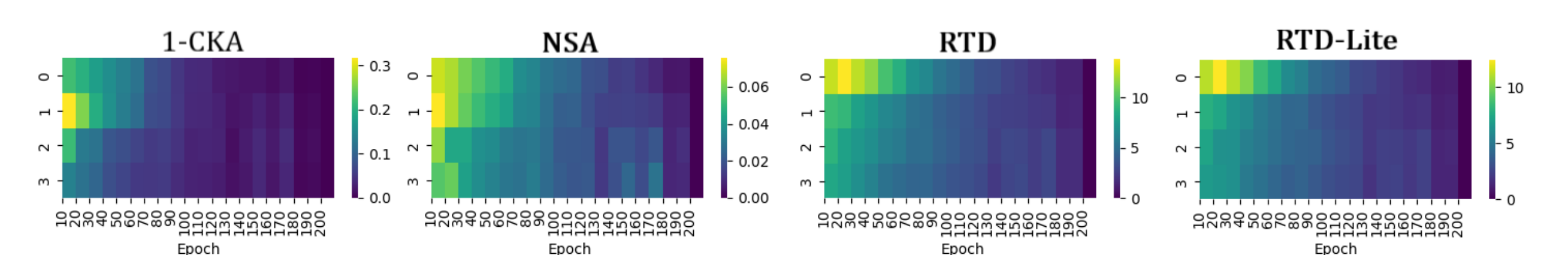


Fig. 4: Evolution of dissimilarity in CGCNs' representations over training epochs.

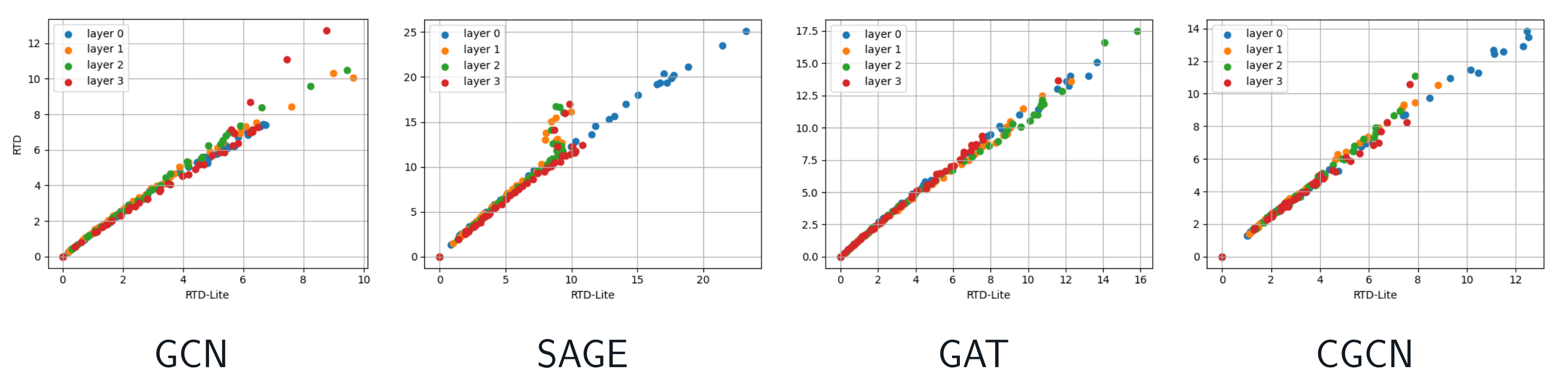


Fig. 5: RTD-Lite vs. RTD at measuring dissimilarity between layer representations in various GNN.

GNN adversarial attacks analysis

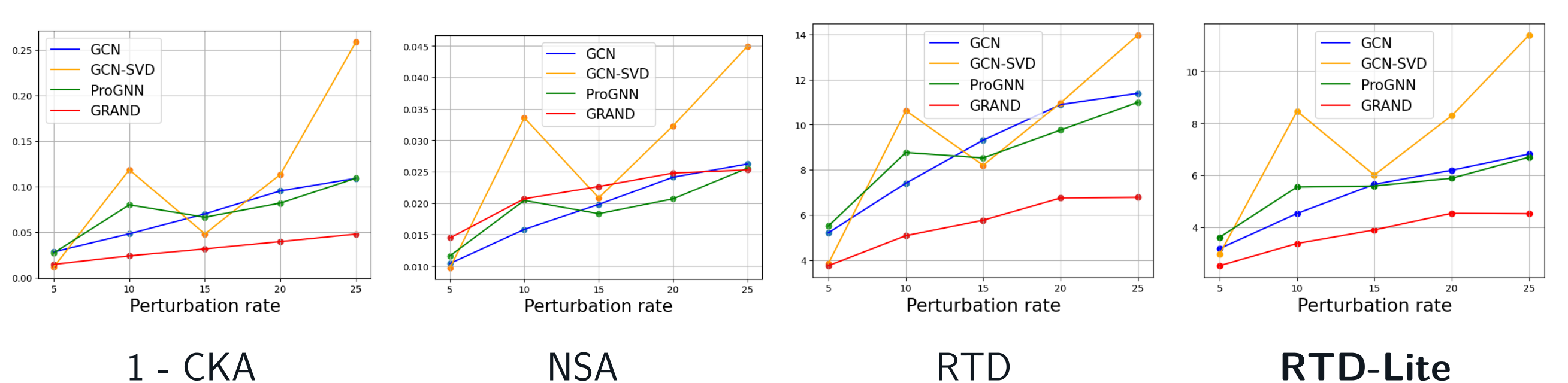


Fig. 6: Dissimilarity in GNNs' representations under Global Evasion attacks.

LLM internal representations analysis

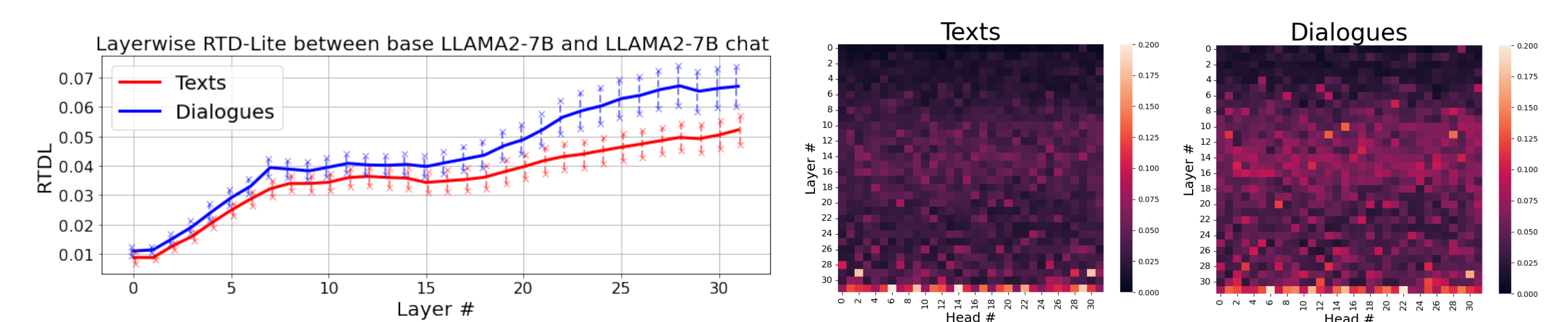


Fig. 7: RTDL between corresponding internal representations of -base and -chat tuned model. Left — layerwise embeddings; Center and Right — attention maps.

Gradient Optimization

- RTD-Lite can be directly minimized with (sub-)gradient methods.
- We trained autoencoder with loss function that included RTD-Lite between dormant (X) and latent (Z) representations:

$$\mathcal{L} = \frac{1}{n} \|X - X_{rec}\|_2^2 + \frac{\lambda}{2n} (RTDL(X, Z) + RTDL(X, Z))$$

Dimensionality reduction quality metric				
Method	Linear Corr.	Wass. dist.	H_0 Triplet Accuracy	RTD
F-MNIST into 16D space				
UMAP	0.602	592.0 ± 3.9	0.741 ± 0.02	12.31 ± 0.4
AE	0.879	320.5 ± 1.9	0.850 ± 0.00	5.52 ± 0.2
TopoAE	0.905	190.7 ± 1.2	0.867 ± 0.01	3.69 ± 0.2
RTD	0.960	181.2 ± 0.8	0.907 ± 0.00	3.01 ± 0.1
RTD-L	0.933	181.7 ± 1.4	0.910 ± 0.01	3.75 ± 0.2
COIL-20 into 16D space				
UMAP	0.301	274.7 ± 0.0	0.574 ± 0.01	15.99 ± 0.5
AE	0.834	183.6 ± 0.0	0.809 ± 0.01	8.35 ± 0.2
TopoAE	0.910	148.0 ± 0.0	0.822 ± 0.02	6.90 ± 0.2
RTD	0.944	88.9 ± 0.0	0.892 ± 0.01	5.78 ± 0.1
RTD-L	0.949	77.1 ± 0.0	0.898 ± 0.01	5.35 ± 0.2

Tab. 1: Quality of data manifold structure preservation at F-MNIST and COIL datasets.